



PARKVISION

Parking Lot Occupancy Detector

YOLOv8 · Computer Vision · Real-Time Detection

The Problem

30%

of urban traffic is caused
by drivers searching for parking

8 min

average time wasted per driver
circling a parking lot

\$97B

lost annually in wasted fuel,
time & productivity

Core Issue: Parking lots have no real-time visibility into which spaces are occupied or available. Drivers must physically search every aisle, creating traffic congestion, fuel waste, frustration, and unnecessary emissions — all of which are entirely preventable with modern computer vision.



Solution Overview



ParkVision uses YOLOv8 to detect occupied & empty spots from overhead camera images in real-time.



INPUT

Overhead parking
lot image (640×640)



DETECT

YOLOv8s runs
object detection



CLASSIFY

Each space labeled
occupied / empty



OUTPUT

Visual overlay +
availability count

✓ **Achieved:** 99.5% mAP@0.5 · 99.8% Precision · 99.8% Recall · ~17ms inference

Technical Approach



Model	YOLOv8s (Ultralytics)	Small model – fast, accurate, pre-trained on COCO
Framework	PyTorch 2.0 + Ultralytics	Industry-standard CV stack
CV Task	Object Detection	Bounding box regression + classification
Input	640 × 640 px images	Overhead parking lot camera frames
Output	Bounding boxes + class labels	occupied / empty with confidence score
Training	Google Colab T4 GPU	50 epochs, early stopping, batch 16

Dataset



Source

PKLot Dataset

Roboflow Universe · sagitova-aliya/pklot-qesrf

~12K

Labeled Images

2

Classes

Dataset Split

Train 70%

Valid 15%

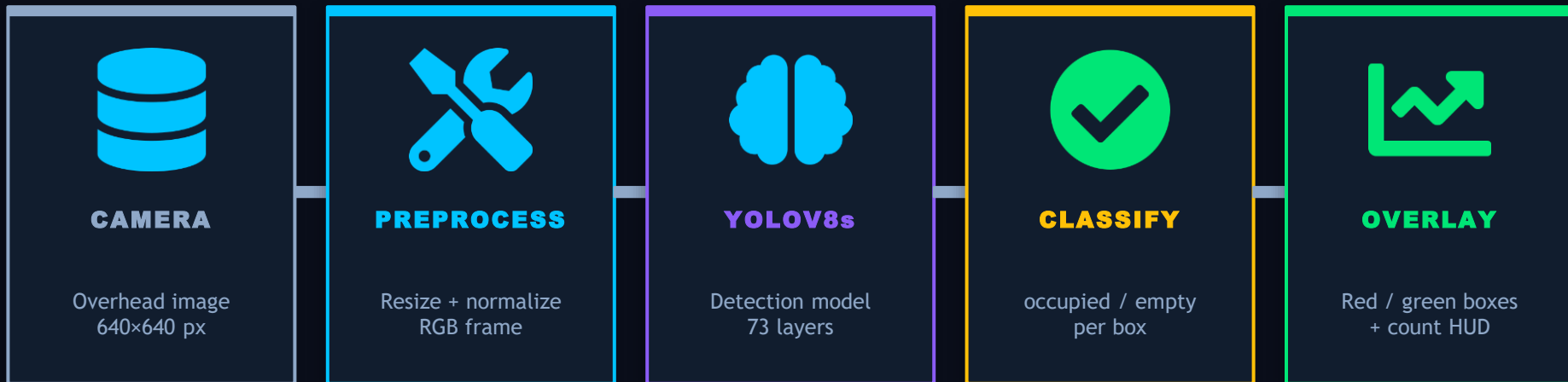
Test 15%

Classes

empty · Available parking space

occupied · Space with a vehicle

System Diagram



Actual Performance

99.5%

mAP@0.5

99.8%

Precision

99.8%

Recall

17ms

Inference

57×

Faster than
target

Results



99.5%

mAP@0.5
(Target $\geq 85\%$)

99.8%

Precision
(Near perfect)

99.8%

Recall
(Near perfect)

17ms

Inference Speed
(Target $< 1000\text{ms}$)

mAP@0.5 Training Curve



Week-by-Week Plan



Week 10	Dataset setup, Colab env, repo structure	Dataset ready	✓
Week 11	Train YOLOv8 baseline model	Model running	✓
Week 12	Fine-tuning, augmentation, validation	99.5% mAP achieved	✓
Week 13	Visual overlay demo (red/green), backup video	Demo notebook ready	
Week 14	Final docs, README update, slide polish	Fully documented	
Week 15	In-class presentation & live demo	Present	

Challenges & Backup Plans



Medium	Occluded / overlapping vehicles → Data augmentation, try YOLOv8m	Mitigated
Medium	Dataset mismatch (different lots) → Supplement with additional Roboflow sets	Mitigated
Low	Google Colab GPU quota limits → Switch to Kaggle (30 hrs/week free GPU)	
Medium	Overfitting on training set → Early stopping + dropout regularization	Mitigated
Low	Live demo failure during class → Pre-recorded backup video of full demo	

Resources & Tools



Compute	Google Colab (T4 GPU)	Free GPU — ~17ms inference	\$0
Model	Ultralytics YOLOv8s	Open-source, Apache 2.0	\$0
Dataset	PKLot via Roboflow Universe	~12,000 labeled images	\$0
Framework	PyTorch 2.0	CUDA acceleration	\$0
IDE	Jupyter / Google Colab	Notebooks committed to GitHub	\$0
Repo	GitHub	YusufShahz/ITAI1378_Midterm_ParkVision	\$0
Total Project Cost: \$0.00 — 100% open-source stack			